B Meenu

Artificial Intelligence

Contact

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Bengaluru, India

Education

REVA University (RACE)

M.Tech. in Artificial Intelligence

Nov 2025 - Nov 2027

REVA University

B.Tech in Artificial Intelligence and Machine Learning

Sep 2022 - Sep 2025

Skills

Technical Skills
Programming Languages: Python, C++, Java, R (RStudio), MATLAB
Artificial Intelligence
Machine Learning, Generative AI, Natural Language Processing (NLP), Computer Vision
Database Management
MySQL, PostgreSQL, Qdrant, Redis
Task-Queue / Async-Processing
Celery (Redis)
Business Intelligence
Power BI, Excel
Soft Skills
Leadership Qualities, Quick Learner, Teamwork, Problem Solving, Critical & Analytical Thinking, Proactive Attitude, Adaptability

Certificates

Python for Data Science(Silver Medalist)-NPTEL
Deep Learning-IIT Ropar (Silver Medalist)-NPTEL
Applied Accelerated Artificial Intelligence-NPTEL
Natural Language Processing-NPTEL Generative AI By Google Cloud
Introduction to Artificial Intelligence MATHLAB for Data Processing and Visualization
Machine Learning using Python
Introduction To Supervised and Unsupervised Machine Learning

About Me

Dynamic AI Developer skilled in Python, TensorFlow, and data analysis. Experienced in prototyping machine learning applications and building data-driven solutions to optimize business processes.Passionate about applying deep learning techniques to solve real-world problems and eager to contribute to innovative AI teams

Projects

AI-Based-Content Moderator for Devanagari Scripts

Personal Project

It involves using advanced Natural Language Processing (NLP) models like mBERT, FastText, and LLMs to detect harmful content (hate speech, threats) and identify targets, focusing on languages like Hindi, Marathi, Nepali, and Sanskrit with a PR-AUC score for fastText is 0.92, while Hate-Speech-CNERG has a score of 0.39 and mBERT stands at 0.78.

Neuro-Fuzzy Autonomous Navigation System

Personal Project

Developed a real-time autonomous navigation system using Interval Type-2 Neuro-Fuzzy Logic (IT2-ANFIS) to process monocular vision data, by modeling environmental uncertainty through a Footprint of Uncertainty (FOU) to ensure robust decision-making in adverse weather and lighting conditions. Engineered a perception pipeline integrating YOLOv8 for lane and obstacle detection with accuracy of 90.45%, while maintaining system explainability by mapping AI decisions to transparent, rule-based telemetry.

Next-Generation RAG: Integrating Corrective Loops, HyDE, and Knowledge Graphs

Personal Project

Engineered a self-healing retrieval system using LangGraph and CRAG (Corrective RAG) that autonomously evaluates document relevance and triggers a Tavily web search fallback to mitigate hallucinations when local data is insufficient.

Enhanced semantic search accuracy by integrating HyDE and Query Decomposition, coupled with a CrossEncoder reranker to ensure the most contextually relevant information is synthesized by the LLM.

Publication and Patent

Multi-Modal Road Hazard Anticipation System using Monocular Vision

Personal Project

- AI-Based-Content Moderator for Devanagari Scripts
- Identification of Different Medicinal Plants/Raw Materials through Image Processing Using Machine Learning Algorithms
- A Cyber Defensive Model for Diabetic Retinopathy Using Bit-Plane Methods